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# Ex - 4

## Implement data mining on non-clinical data to perform chronic disease management.

## Aim :

This Exercise aims to perform data exploration and clustering on a set of non-clinical data features, with visualization at key stages to help understand the data.
Data mining is used to analyze and uncover patterns in this data.and

## Objective :

This Exercise aims to perform data exploration and clustering on a set of non-clinical data features, with visualization at key stages to help understand the data.
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## Procedure :

### Medical Text Analysis Workflow

1. **Set Up**: Initialize Jupyter notebook and create a new notebook.
2. **Import Libraries**: Import necessary libraries:
   - `pandas`, `numpy` (Data handling)
   - `re`, `nltk`, `spacy` (NLP & Cleaning)
   - `transformers` (Hugging Face models)
   - `wordcloud`, `matplotlib` (Visualization)
3. **Load Data**: Load medical text data.
4. **Preprocess Text Data**: Clean and tokenize the text data.
5. **Named Entity Recognition (NER)**: Use **spaCy** to perform NER on the medical text data.
6. **Sentiment Analysis**: Use the **transformers** library to perform sentiment analysis.
7. **Generate Word Cloud**: Create a word cloud to visualize the most common terms.
8. **Extract Insights**: Combine extracted entities and sentiment results to generate patient insights.

# Step 0 : Import Necessary Libraries

Generate + Code + Markdown

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
```

[21] Python

- **Explanation :**

- **Pandas** : Data manipulation and analysis (often used for working with DataFrames/Tabular Data).
- **NumPy** : Numerical computations, used to generate random data and perform mathematical operations.
- **Matplotlib.pyplot** : A plotting library used to create graphs and static visualizations.
- **Seaborn** : A data visualization library built on top of Matplotlib, offering more advanced and aesthetically pleasing plots.
- **StandardScaler** : A preprocessing tool to standardize data so that each feature has a mean of $0$ and a standard deviation of $1$.
- **PCA (Principal Component Analysis)** : A technique used to reduce the dimensionality of the data while retaining as much information (variance) as possible.
- **KMeans** : A popular clustering algorithm used to group data into $k$ number of clusters based on similarity.

# Step 1 : Load the Data(creating the Data)

```python
data = {
    'Age' : np.random.randint(20,80,100),
    'BMI' : np.random.randint(19,35,100),
    'Smoking' : np.random.randint(0,2,100),
    'AlcoholConsumption' : np.random.randint(0,4,100),
    'PhysicalActivity' : np.random.randint(0,4,100),
    'ChronicDisease' : np.random.randint(0,2,100)
}
```

[22] Python

We define a dictionary where each **key** is a feature and each **value** is an array of random integers generated via `np.random.randint()`.

## Data Dictionary Breakdown:

- **Age**: Random integers between $20$ and $80$ (100 samples).
- **BMI**: Random integers between $18$ and $35$ (Body Mass Index).
- **Smoking**: Binary values ($0$ or $1$) representing smoking status.
- **AlcoholConsumption**: Levels from $0$ to $3$.
- **PhysicalActivity**: Activity levels ($0, 1, 2$).
- **ChronicDisease**: Binary values ($0$ or $1$) indicating presence of disease.

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## Converting the Data into a DataFrame

[23]

```python
non_clinical_data = pd.DataFrame(data)
```

Python

The dictionary data is converted into a pandas DataFrame named non_clinical_data.

### Displaying the First Few Rows

[24]

```python
(non_clinical_data.head(7))
```

Python

| | Age | BMI | Smoking | AlcoholConsumption | PhysicalActivity | ChronicDiseas |
|---|---|---|---|---|---|---|
| 0 | 26 | 22 | 0 | 2 | 0 | |
| 1 | 58 | 21 | 1 | 0 | 1 | |
| 2 | 51 | 23 | 0 | 1 | 0 | |
| 3 | 43 | 25 | 1 | 1 | 2 | |
| 4 | 72 | 26 | 1 | 2 | 3 | |
| 5 | 36 | 27 | 1 | 3 | 1 | |
| 6 | 54 | 20 | 0 | 3 | 0 | |

```python
non_clinical_data.describe()
```

[25] Python

| | Age | BMI | Smoking | AlcoholConsumption | PhysicalActi |
|---|---|---|---|---|---|
| count | 100.000000 | 100.000000 | 100.000000 | 100.000000 | 100.000 |
| mean | 46.120000 | 25.880000 | 0.480000 | 1.320000 | 1.500 |
| std | 17.676258 | 4.573264 | 0.502117 | 1.081338 | 1.105 |
| min | 20.000000 | 19.000000 | 0.000000 | 0.000000 | 0.000 |
| 25% | 30.000000 | 22.750000 | 0.000000 | 0.000000 | 0.750 |
| 50% | 43.500000 | 26.000000 | 0.000000 | 1.000000 | 2.000 |
| 75% | 60.250000 | 30.000000 | 1.000000 | 2.000000 | 2.000 |
| max | 79.000000 | 34.000000 | 1.000000 | 3.000000 | 3.000 |

# Step 2 : Preprocessing of Data

```python
# Check for missing values
non_clinical_data.isnull().sum()
```

[26] Python

```
Age                  0
BMI                  0
Smoking              0
AlcoholConsumption   0
PhysicalActivity     0
ChronicDisease       0
```

# Standardise The Features

```
Features = ['Age','BMI','Smoking','AlcoholConsumption','PhysicalActiv

scaler = StandardScaler()
non_clinical_data[Features] = scaler.fit_transform(non_clinical_data[
```

[27] Python

- **Process : Data Standardization**

- **Method** : The selected features (**Age**, **BMI**, **Smoking**, **AlcoholConsumption**, and **PhysicalActivity**) are processed using `StandardScaler`.
- **Function** : This transforms each feature to have a mean of 0 and a standard deviation of 1.
- **Importance** : This step is crucial for distance-based algorithms like **PCA** and **KMeans**, which are sensitive to the scale of the data (e.g., ensuring "Age" values don't outweigh "Smoking" values just because the numbers are larger).

```python
non_clinical_data.head()
```

[28]

Python

| | Age | BMI | Smoking | AlcoholConsumption | PhysicalActivity | Ch |
|---|---|---|---|---|---|---|
| 0 | -1.143984 | -0.852683 | -0.960769 | 0.632018 | -1.363636 | |
| 1 | 0.675474 | -1.072447 | 1.040833 | -1.226859 | -0.454545 | |
| 2 | 0.277467 | -0.632920 | -0.960769 | -0.297420 | -1.363636 | |
| 3 | -0.177397 | -0.193392 | 1.040833 | -0.297420 | 0.454545 | |
| 4 | 1.471487 | 0.026372 | 1.040833 | 0.632018 | 1.363636 | |

```python
non_clinical_data.describe()
```

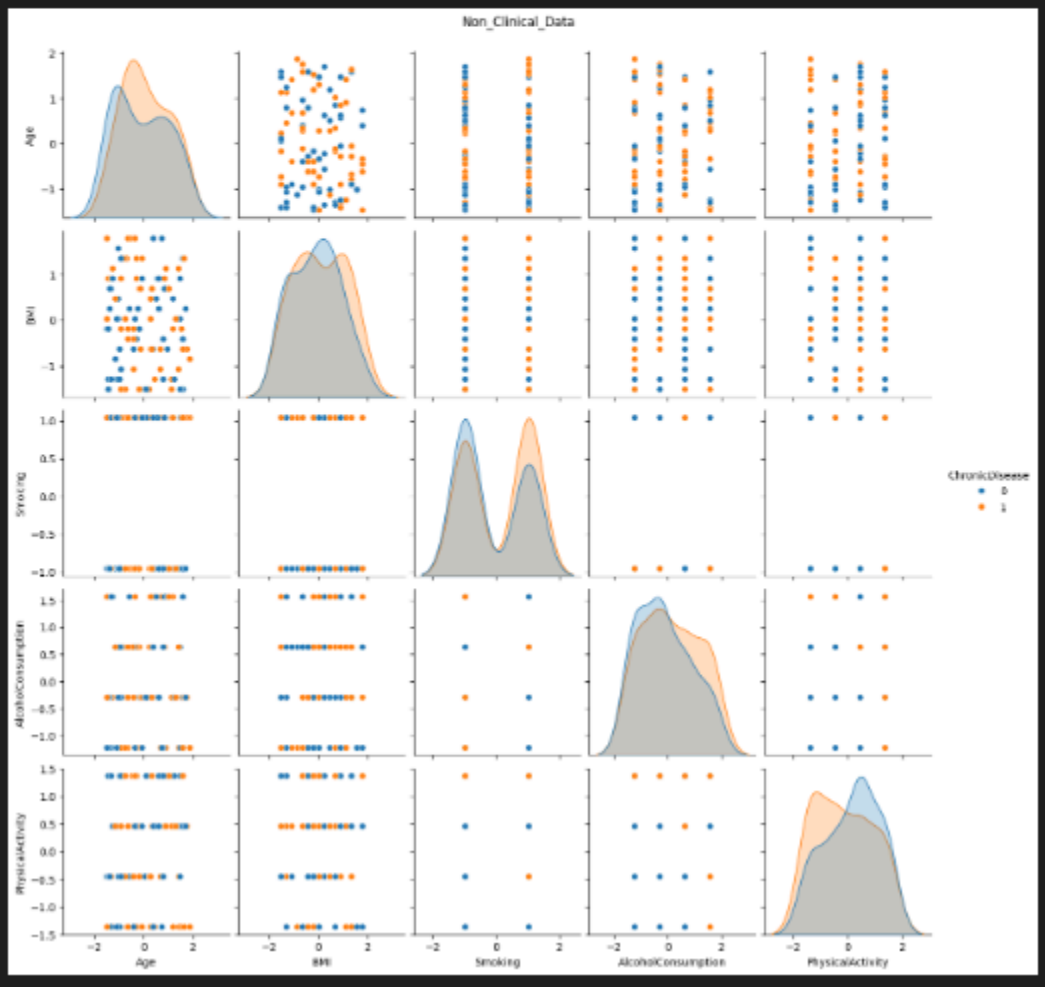
Python

| | Age | BMI | Smoking | AlcoholConsumption | Ph |
|---|---|---|---|---|---|
| count | 1.000000e+02 | 1.000000e+02 | 1.000000e+02 | 1.000000e+02 | |
| mean | 1.376677e-16 | 1.962666e-16 | -2.220446e-17 | -3.996803e-17 | |
| std | 1.005038e+00 | 1.005038e+00 | 1.005038e+00 | 1.005038e+00 | |
| min | -1.485133e+00 | -1.511975e+00 | -9.607689e-01 | -1.226859e+00 | - |
| 25% | -9.165520e-01 | -6.878606e-01 | -9.607689e-01 | -1.226859e+00 | - |
| 50% | -1.489681e-01 | 2.637165e-02 | -9.607689e-01 | -2.974204e-01 | |
| 75% | 8.034044e-01 | 9.054267e-01 | 1.040833e+00 | 6.320183e-01 | |
| max | 1.869493e+00 | 1.784482e+00 | 1.040833e+00 | 1.561457e+00 | |

# Step 3 : Visualize the Data using PairPlot

```python
sns.pairplot(non_clinical_data,hue='ChronicDisease')
plt.suptitle('Non_Clinical_Data',y=1.02)
plt.show()
```

[30] Python



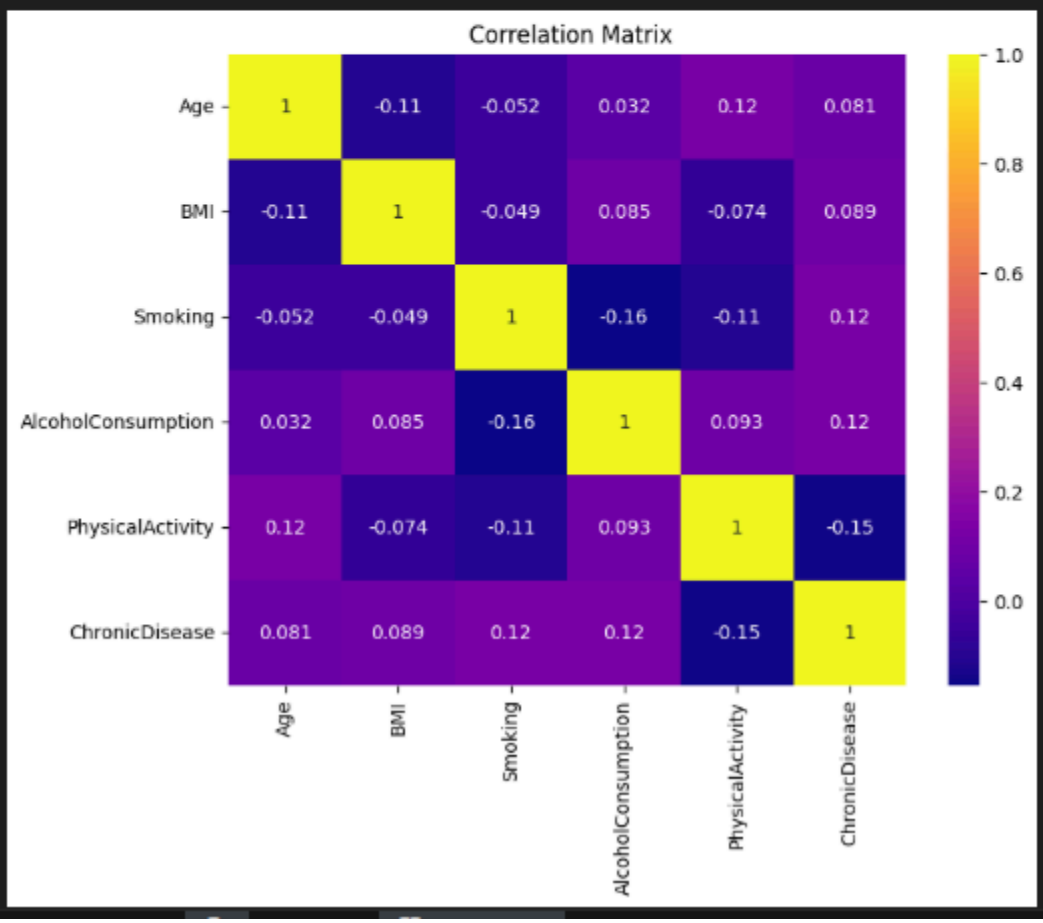
- **Visualization : Pairplot**

- **Definition** : This creates a grid of scatter plots for every pair of features in the dataset.
- **Color-Coding** : Data points are colored based on the **ChronicDisease** column (the target variable).
- **Purpose** : It provides an intuitive way to visually explore relationships between features and helps identify patterns or clusters associated with chronic disease.

# Step 4 : Plotting a Correlation HeatMap

```python
plt.figure(figsize=(8,6))
sns.heatmap(non_clinical_data.corr(), annot=True, cmap='plasma')
plt.title(" Correlation Matrix")
plt.show()
```

[31] Python



- **Visualization : Correlation Heatmap**

- **Definition** : This matrix displays how strongly the features are correlated with each other, with values ranging from $-1$ (perfect negative correlation) to $1$ (perfect positive correlation).
- `annot=True` : This parameter overlays the actual numerical correlation values on the heatmap for better readability.
- `cmap='coolwarm'` : This sets the color palette; typically, "cool" (blue) represents negative correlation and "warm" (red) represents positive correlation.

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# Step 5 : Applying PCA (for dimentionality reduction)

```python
pca = PCA(n_components=2)
principal_componets = pca.fit_transform(non_clinical_data[Features])
pca_df = pd.DataFrame(principal_componets, columns=['PC1','PC2'])
pca_df['ChronicDisease'] = non_clinical_data['ChronicDisease']
```

[32] Python

# Step 5 : Applying PCA (for dimentionality reduction)

```python
pca = PCA(n_components=2)
principal_componets = pca.fit_transform(non_clinical_data[Features])
pca_df = pd.DataFrame(principal_componets, columns=['PC1','PC2'])
pca_df['ChronicDisease'] = non_clinical_data['ChronicDisease']
```

[32] Python

- **Process : Principal Component Analysis (PCA)**

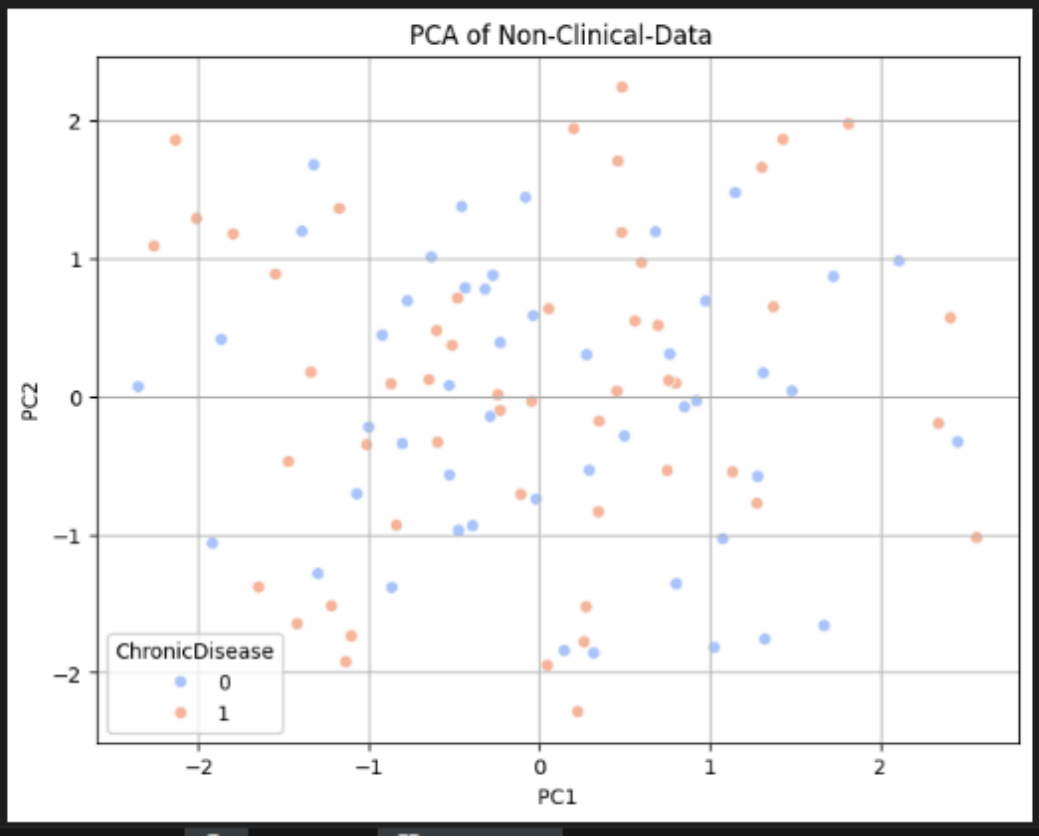
- **Dimensionality Reduction** : PCA reduces the dataset from **5 features** down to **2 principal components** (PC1 and PC2) while retaining as much information (variance) as possible.
- `fit_transform()` : This method calculates the components and performs the transformation, storing the result in `principal_components`.
- `pca_df` : A new DataFrame is created containing the two components and the **ChronicDisease** column, making it possible to visualize a 5-dimensional problem in a simple 2D scatter plot.

# Step 6 : Visualizing the PCA Results

```python
plt.figure(figsize=(8, 6))
sns.scatterplot(x='PC1', y='PC2', hue='ChronicDisease', data=pca_df,

plt.title(" PCA of Non-Clinical-Data")
plt.xlabel(' PC1 ')
plt.ylabel(' PC2 ')
plt.grid(True)
plt.show()
```

[33] Python



- **Visualization : PCA Scatter Plot**

- **Definition** : This visualizes the data in a new 2-dimensional space created by the Principal Components.
- **Color-Coding** : Each point is colored based on the **ChronicDisease** status, allowing for easy observation of group separation.
- **Purpose** : It helps to determine how well the non-clinical data can be separated or clustered in a reduced-dimensional space to identify potential health patterns.

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# Step : 7 Applying KMeans Clustering

kmeans = KMeans(n_clusters=2, random_state=0) non_clinical_data['Cluster'] = kmeans.fit_predict(non_clinical_data[Features]) pca_df['Cluster'] = non_clinical_data['Cluster']

- **Algorithm : KMeans Clustering**

- **Grouping** : `KMeans` is used to partition the data into **n_clusters=2**, aligning with the two expected groups (e.g., Higher Risk vs. Lower Risk).
- **Cluster Labels** : The algorithm assigns each data point to one of the two clusters based on feature similarity; these assignments are stored in a new **Cluster** column.
- **Integration** : These labels are added to the **pca_df** DataFrame, allowing us to visualize how the mathematical clusters overlap with the actual **ChronicDisease** status.

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# Step 8 : Visualizing the KMeans Clusters

```python
plt.figure(figsize=(10,8))
sns.scatterplot(x='PC1', y='PC2', hue='Cluster', data=pca_df,palette=

plt.title("KMeans clustering of Non Clinical Data")
plt.xlabel(' PC1 ')
plt.ylabel(' PC2 ')
plt.grid(True)
plt.show()
```

[35] Python

## 🧠 Sentiment Analysis

### Need:

- **Tone Identification**: Detects the underlying emotional tone within medical records and clinical narratives.
- **Psychological Insight**: Helps clinicians understand the patient's psychological state and overall outlook on their health.
- **Urgency Detection**: Flags high levels of concern or urgency in clinical notes that may require immediate follow-up.
- **Context Enrichment**: Complements objective clinical data with subjective emotional context for a holistic patient view.

### Inference:

Even clinical texts carry emotional undertones. Sentiment analysis reveals critical indicators of **patient well-being** and **treatment satisfaction**, highlighting areas that may need additional support beyond physical care.

- **Visualization : Scatter Plot of Clusters**

- **Definition** : This plot displays the results of the **KMeans** algorithm within the 2D space generated by PCA.
- **Color-Coding** : Each data point is colored according to its assigned **Cluster** (e.g., Cluster 0 vs. Cluster 1).
- **Purpose** : It provides a visual confirmation of how the algorithm has partitioned the individuals into distinct groups based on their non-clinical risk factors.

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# Final Inferences and Analysis

## 1. Interpreting the Cluster Alignment

- **Successful Identification**: If the clusters align well with the **ChronicDisease** groups (e.g., Cluster 1 contains mostly individuals with the disease), it suggests that KMeans has identified meaningful biological and lifestyle patterns related to health outcomes.
- **Weak Alignment**: If the clusters overlap significantly or do not match the disease labels, it suggests that the current non-clinical features alone may not be enough to capture the underlying health structure, or the relationship is non-linear.

## 2. Summary of the Analysis (Exercise 4)

- **Dimensionality Reduction**: **PCA** successfully condensed the 5D dataset into two principal components. The resulting scatter plot reveals whether non-clinical factors—such as **Age**, **BMI**, and **Physical Activity**—are sufficient to visually separate healthy individuals from those at risk.
- **Predictive Potential**: The **KMeans** clustering results demonstrate our ability to group individuals based on lifestyle similarities. Strong alignment between these clusters and actual disease labels indicates that **data mining techniques** are a powerful tool for predicting chronic disease risk using only non-clinical data.

## ☁️ Word Cloud Visualization

### Need:

- **Term Importance**: Provides a visual representation where the size of the text indicates the frequency and importance of specific terms.
- **Theme Identification**: Allows for the quick identification of dominant medical themes across the entire dataset.
- **Highlight Focus**: Surfaces frequently mentioned conditions and treatments at a single glance.
- **Dataset Validation**: Serves as a sanity check to ensure the data aligns with the intended medical domain.

### Inference:

The word cloud serves as a visual confirmation of our focus; it should prominently feature terms like **hypertension**, **diabetes**, and **asthma**, validating that the preprocessing and data loading were successful.

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